

Lean Formalization of the Parameter Classification for Conditional Entropies

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This note explains how Lean 4 verifies the necessary and sufficient parameter conditions for the concrete conditional-entropy candidates studied in the complete-proof paper, Ref. [1]. The formalization covers all thirty-one numbered results: sixteen results in Sections 4 and 5 and fifteen technical results in Appendix A. For each result, the note identifies one distinct Lean theorem with the same mathematical content, allowing the reader to move directly from a statement in the paper to its formal proof. A dependency diagram also shows how the main classification is built from the underlying calculus, measure-theoretic, localization, and semiring/channel results. Every supporting result is proved using Lean and Mathlib: the development contains no unfinished proofs, project-defined axioms, or additional unproved assumptions.

1. LEAN CORRESPONDENCE

The formal development follows the complete analytic proofs in Ref. [1]. It covers the endpoint-aware Rényi parameter, real-power and entropy differentiation, signed integration, common discretization, the two-block, three-block, and Shannon localizers, exact tropical stationarity, all sufficiency and necessity branches, and the original finite semiring and conditionally mixing channel lift. The source is available at <https://github.com/robertorubboli/A-complete-characterization-of-conditional-entropies> on the repository's main branch.

A Lean file contains definitions and named mathematical statements together with their proofs. Lean calls each named item a *declaration*. A theorem declaration records exactly which assumptions may be used and which conclusion must be proved. When the project is built, Lean checks every step using Mathlib, its standard mathematical library, and declarations that have already been proved in the project. This project rejects unfinished-proof markers, and Lean accepts each remaining theorem only when its proof checks.

Every numbered lemma, proposition, and remark in the complete-proof paper, Ref. [1], has one matching Lean theorem. The theorem's name finds its complete formal statement and the proof that combines any required supporting lemmas. For example, Lemma 4.1 is represented by `ConditionalEntropy.fullDetailsLemma4_1`. A theorem may use several smaller proved lemmas internally, just as a written proof may refer to earlier results.

The formal proof ends with two final theorems. The theorem `mainClassification` states that, for the given probability measure on the compactified Rényi parameter space, each concrete candidate is monotone exactly when its parameter satisfies the stated admissibility condition. Its only explicit input is the measure being classified; it needs no additional hypothesis. The theorem `allConditionalEntropyAxioms` proves nonnegativity, zero-embedding invariance, tensor additivity, fair-bit normalization, and embedded channel monotonicity for every admissible candidate, with each admissibility condition written explicitly. It assumes no additional fact beyond those stated conditions. Neither theorem uses a project-defined axiom. In the complete-proof paper and in Lean, $\log$ is the natural logarithm and an independent fair bit has value $\log 2$.

The two dependency diagrams show the proof at different levels of detail. Figure 1 is a short overview. Starting at either green box and following the arrows downward shows which results are used in that proof. White boxes represent numbered results from the complete-proof paper, while blue boxes collect smaller supporting results proved in Lean.

The files `paper/dependency-graph.dot`, `paper/dependency-graph.svg`, and `paper/dependency-graph.png` are editable, vector-image, and bitmap-image versions of one more detailed diagram. That diagram shows the individual Lean dependencies. Its arrows point from an earlier result to the later result that uses it, which is the opposite direction from the arrows in Figure 1.

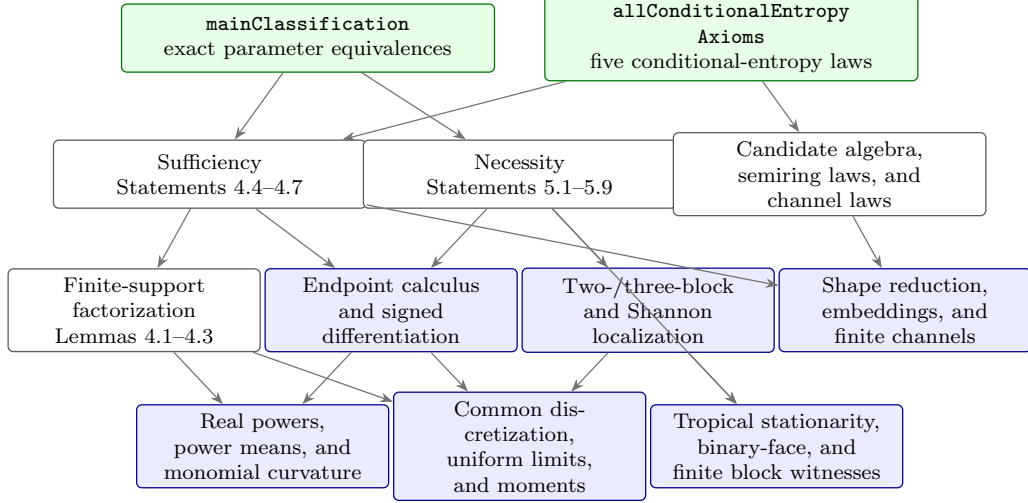


FIG. 1. Short proof map for the parameter classification. An arrow from one box to another means that the first result uses the second in its proof. Green boxes are the two final theorems, white boxes are numbered results from the complete-proof paper, Ref. [1], and blue boxes are supporting results proved in Lean. The detailed implementation graph uses the opposite arrow direction: from each proved ingredient to the result that it supports.

Table I connects each numbered paper result with its matching Lean theorem. The columns give the manuscript result, the exact Lean name, and the status. “Exact” means that the whole mathematical statement has been proved, not only a special case or one part of it.

The sixteen results in Sections 4 and 5, including Remarks 5.4–5.5, have sixteen different matching theorems. Their names begin with `ConditionalEntropy` and they are defined in `ConditionalEntropy/FullDetailsStatements.lean`. Inside its proof, a matching theorem may call smaller proved lemmas. The source therefore has more names than the table, but the correspondence remains one-to-one: the table lists exactly one complete theorem for each numbered manuscript result, and the dependency graphs show the additional proof steps.

TABLE I. Statement-by-statement correspondence. All numbers are from the complete-proof paper, Ref. [1].

Manuscript statement	Lean declaration	Status
Lemma 4.1: Power-mean curvature	fullDetailsLemma4_1	Exact
Lemma 4.2: Monomial curvature	fullDetailsLemma4_2	Exact
Lemma 4.3: Finite-support sufficiency	fullDetailsLemma4_3	Exact
Proposition 4.4: General temperate sufficiency	fullDetailsProposition4_4	Exact
Proposition 4.5: Negative-tropical sufficiency	fullDetailsProposition4_5	Exact
Proposition 4.6: Positive-tropical sufficiency	fullDetailsProposition4_6	Exact

Table I (continued; complete-proof numbering).

Manuscript statement	Lean declaration	Status
Proposition 4.7: Derivation sufficiency	fullDetailsProposition4_7	Exact
Proposition 5.1: Positive necessity	fullDetailsProposition5_1	Exact
Proposition 5.2: Negative Shannon atom excludes upper support	fullDetailsProposition5_2	Exact
Proposition 5.3: Truncated exceptional-moment bound	fullDetailsProposition5_3	Exact
Remark 5.4: Unnormalised cone witnesses and homogeneity	fullDetailsRemark5_4	Exact
Remark 5.5: Compensated two-column witness	fullDetailsRemark5_5	Exact
Proposition 5.6: Exceptional negative branch: unique upper point, finite one-sided moments, and moment bound	fullDetailsProposition5_6	Exact
Proposition 5.7: Negative-tropical necessity	fullDetailsProposition5_7	Exact
Proposition 5.8: Positive-tropical necessity	fullDetailsProposition5_8	Exact
Proposition 5.9: Derivation necessity	fullDetailsProposition5_9	Exact

Table II applies the same rule to the fifteen numbered results in Appendix A. Together, the two tables give all thirty-one paper-to-Lean pairs: sixteen from Sections 4 and 5 and fifteen from Appendix A.

The file `paper/full-details-correspondence.json` records the same thirty-one pairs in a form that can be checked automatically. The separate 152-row blueprint lists labeled definitions and results used to construct the formalization, including supporting and final results. These rows describe the internal structure and do not correspond to additional numbered manuscript statements. The detailed dependency graph shows how the entries are connected.

The theorem for Lemma A.9 includes both parts of that lemma: continuity along the entropy line and the exact cancellation integral for orders zero, one, and two. Thus every row in both tables represents the complete stated result.

TABLE II. Appendix statement correspondence. All numbers are from Appendix A of the complete-proof paper, Ref. [1].

Manuscript statement	Lean declaration	Status
Lemma A.1: Hessian of the power mean	<code>fullDetailsAppendixA_1</code>	Exact
Lemma A.2: Hessian and sign classification of a monomial	<code>fullDetailsAppendixA_2</code>	Exact
Lemma A.3: Continuity and uniform bound for finite-dimensional Rényi entropy	<code>fullDetailsAppendixA_3</code>	Exact
Lemma A.4: Concavity and Schur concavity in the sufficient-condition ranges	<code>fullDetailsAppendixA_4</code>	Exact
Lemma A.5: Measure-theoretic tools	<code>fullDetailsAppendixA_5</code>	Exact
Lemma A.6: Common discrete approximation of the Rényi parameter	<code>fullDetailsAppendixA_6</code>	Exact
Lemma A.7: Pointwise limits preserve the shape inequalities	<code>fullDetailsAppendixA_7</code>	Exact
Lemma A.8: First and second derivatives of the logarithmic power mean	<code>fullDetailsAppendixA_8</code>	Exact

Table II (continued).

Manuscript statement	Lean declaration	Status
Lemma A.9: Cancellation and continuity through the Shannon point	<code>fullDetailsAppendixA_9</code>	Exact
Lemma A.10: Continuity of the differentiated entropy at the parameter endpoints	<code>fullDetailsAppendixA_10</code>	Exact
Proposition A.11: Differentiation under the Rényi-parameter integral at fixed dimension	<code>fullDetailsAppendixA_11</code>	Exact
Lemma A.12: Dominant-block remainder	<code>fullDetailsAppendixA_12</code>	Exact
Lemma A.13: First two derivatives of Shannon entropy along an affine path	<code>fullDetailsAppendixA_13</code>	Exact
Lemma A.14: Second-order condition for a one-dimensional quasi-convex function	<code>fullDetailsAppendixA_14</code>	Exact
Lemma A.15: Exact finite-dimensional stationarity correction	<code>fullDetailsAppendixA_15</code>	Exact

A. What is checked

The project uses Lean 4.32.2 and Mathlib 4.32.2. Its source contains no `sorry` or `admit`, which are markers for unfinished proofs, and no project-defined `axiom`. It also uses no project-defined `opaque` declaration. Thus none of the manuscript results has been inserted into the project as an unproved fact. The final theorems require no extra project hypothesis. Like other Lean developments based on Mathlib, the project uses Lean’s standard logical foundations, including classical choice, propositional extensionality, and quotient soundness.

The command

```
powershell -File scripts/check.ps1
```

performs the complete check. It rejects unfinished-proof markers, builds all Lean files with warnings treated as errors, reports the standard logical foundations used by the final theorems, and checks the 152 labeled definitions and results recorded in the blueprint. It also reads all thirty-one result numbers and labels from the LaTeX paper, verifies that their Lean names are distinct and appear in the same order, and asks Lean to check every corresponding theorem. A successful run therefore checks both the proofs and the paper-to-Lean correspondence. The formalization covers the classification and the five entropy laws for all four concrete candidate families, using the natural-log convention of the complete-proof paper.

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- [1] R. Rubboli, E. Haapasalo, and M. Tomamichel. “*Sufficient and Necessary Conditions for the Parameters of Conditional Entropies: Complete Proofs*”. Auxiliary document on GitHub, (2026). Available online: <https://github.com/robertorubboli/A-complete-characterization-of-conditional-entropies/blob/main/paper/auxiliary-files/pdf/sections-4-5-full-details.pdf>. Complete proofs for Sections 4 and 5 of the canonical manuscript.